

The Alaoglu–Erdős Integer-Exponent Conjecture

Further Determinant Methods,
Arbitrary-Node Repulsion,
and the Remaining Four-Exponentials Barrier

A repository-grounded research report based on

<https://github.com/VladimirReshetnikov/ProveIt>

pinned at commit

5d3982ceb12644fdd3499569d9db5f62f9a31dc6

Bottom line. No unconditional proof of the conjecture is obtained. The current ProveIt corpus already pushes the conditional structure remarkably far: if a counterexample exists, then all solutions form the free additive monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$ generated by 1 and one least nonintegral exponent β , and all natural candidate bases form the free multiplicative monoid generated by 2 and $M = 2^\beta$.

This report adds two paper-level determinant methods. First, arbitrary candidate tuples on the lattice curve $y = x^\vartheta$ satisfy an explicit Vandermonde repulsion inequality; this strictly extends the repository’s arithmetic-progression obstructions to completely irregular nodes. Second, the integrality of every mixed monomial $m^{a+b\vartheta} = m^a(m^\vartheta)^b$ gives a hierarchy of positive integer generalized Vandermonde determinants. It yields an independent, explicit $O((\log X)^2)$ bound for candidates in $[X, 2X]$, without using the prime-valuation rank theorem. The method also diagnoses its own failure: a hypothetical counterexample supplies only $\asymp \log X$ points in such a block, while the archimedean determinant needs $\asymp (\log X)^2$ points to become nearly scale-forcing. Closing precisely this one-logarithm deficit appears to require a genuinely new ingredient, most plausibly p -adic divisibility, a denominator-sensitive counting theorem, or a new algebraic exponential.

August 20, 2026

Abstract

Let $\vartheta = \log 3 / \log 2$. The Alaoglu–Erdős conjecture is equivalent to the assertion that the only positive integers m for which m^ϑ is an integer are the powers of 2. The pinned ProveIt development contains kernel-checked zero-free regions, transcendence restrictions, multiplicative-rank and density bounds, a canonical primitive odd core, a localized-radical reformulation, exact radical degree, arbitrary-order arithmetic-progression obstructions, and — conditional on failure — a complete classification

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}$$

with unique coordinates. It also gives exact dyadic-block counts and a sharp quadratic cumulative asymptotic. These results reduce the conjecture to a very narrow special case of the four exponentials problem: no primitive odd $w > 1$ should have a positive power of w^ϑ in $\mathbb{Z}[1/3]$.

We attempt several routes to the remaining obstruction and make their failure points explicit. The principal new theorem is an arbitrary-node interpolation determinant. If $1 \leq m_0 < \dots < m_r$ and every m_i^ϑ is integral, then for $r \geq 2$

$$\prod_{0 \leq i < j \leq r} (m_j - m_i) \geq \frac{r!}{|(\vartheta)_r|} m_0^{r-\vartheta}.$$

For three candidates in $[N, N + H]$ this gives

$$H \geq \left(\frac{8}{\vartheta(\vartheta - 1)} \right)^{1/3} N^{(2-\vartheta)/3} = 2.05107 \dots N^{0.138345 \dots}.$$

The order-4 member gives the best exponent among the ordinary-power family: every interval shorter than $1.46022 N^{0.241503 \dots}$ contains at most four candidates.

A stronger hierarchy uses the ordered semigroup $\Lambda_\vartheta = \{a + b\vartheta : a, b \in \mathbb{N}_0\}$. For its first $r + 1$ exponents $\lambda_0 < \dots < \lambda_r$, every candidate tuple produces the strictly positive integer

$$\det(m_i^{\lambda_j})_{0 \leq i, j \leq r}.$$

A divided-difference estimate gives, with $R = r(r + 1)/2$ and $\Sigma_r = \sum_{j=0}^r \lambda_j$,

$$H \geq ((r + 1)!)^{-1/R} 2^{-1-2\Sigma_r/R} N^{1-\Sigma_r/R}$$

whenever $r + 1$ candidates lie in $[N, N + H] \subseteq [N, 2N]$. Since $\Sigma_r/R = (4/3)\sqrt{2\vartheta/r} + O(r^{-1})$, the exponent tends to 1. An entirely explicit corollary is

$$\#(\mathcal{C} \cap [X, 2X]) \leq 10000 \max\{1, \log X\}^2.$$

This is independent of the repository’s valuation-rank proof and is suitable for Lean formalization. It still does not contradict the exact conditional monoid, whose block count is only $\asymp \log X$. The report ends with new equivalent formulations, a prioritized research program, and a module-by-module formalization blueprint.

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1 Executive verdict

1.1 What was attempted

The investigation began from the current combined report and the Lean modules imported by `ExponentialIdentities.lean` at commit `5d3982ceb12644fdd3499569d9db5f62f9a31dc6`. The main lines pursued were:

1. an exact reduction to the integer curve $m \mapsto m^\vartheta$ and to the logarithmic 2×2 determinant behind four exponentials;
2. attempts to combine the canonical primitive-generator theorem with algebraic-number and radical-field arguments;
3. rational approximation to a hypothetical least generator, including simultaneous integer lower bounds at bases 2 and 3;
4. density and shrinking-target arguments for the fractional parts of its multiples;
5. finite differences beyond arithmetic progressions;
6. interpolation determinants on arbitrary candidate nodes;
7. generalized Vandermonde determinants using the full semigroup of integral mixed monomials $m^{a+b\vartheta}$;
8. possible p -adic, S -integral, and auxiliary-base amplifications.

The first five routes reach known barriers. The sixth and seventh produce new unconditional partial results. None closes the final transcendence-theoretic gap.

1.2 The strongest new conclusions

New theorem A: arbitrary-node repulsion. For every $r \geq 2$, any $r + 1$ distinct natural candidates obey a quantitative product-of-gaps lower bound. No arithmetic-progression hypothesis is needed. This gives explicit local packing bounds and a best ordinary-power exponent $0.241503 \dots$ at determinant order 4.

New theorem B: semigroup determinant hierarchy. The mixed exponent semigroup $\mathbb{N}_0 + \mathbb{N}_0\vartheta$ supplies arbitrarily large positive integer generalized Vandermonde determinants. Their archimedean upper bounds yield a near-linear repulsion exponent $1 - O(r^{-1/2})$ and, unconditionally,

$$\#(\mathcal{C} \cap [X, 2X]) \ll (\log X)^2.$$

An explicit constant 10000 is proved below. This route is independent of the repository’s prime-valuation rank theorem.

New diagnosis: the one-logarithm deficit. To make the semigroup determinant nearly force a gap comparable with X , one needs $r \asymp (\log X)^2$ candidate nodes. Conditional on a counterexample, the exact ProveIt classification supplies only $\asymp \log X$ candidates in $[X, 2X]$. Thus purely archimedean interpolation, at its present strength, misses the conditional population by one power of $\log X$. A successful refinement must either reduce the determinant’s node requirement to $O(\log X)$ or extract additional nonarchimedean size from the same nodes.

1.3 Why this does not amount to a proof

The conjecture is already a special integral instance of the real four exponentials conjecture. In the natural 2×2 matrix

$$\begin{pmatrix} 1 \cdot \log 2 & 1 \cdot \log 3 \\ x \log 2 & x \log 3 \end{pmatrix},$$

the four exponentials are $2, 3, 2^x, 3^x$. For a nonintegral solution, both row and column pairs are rationally linearly independent, yet all four exponentials are algebraic. The conclusion needed here is therefore exactly what four exponentials predicts and what the known six exponentials theorem does not supply in a 2×2 configuration.

The additional integrality hypotheses are stronger than mere algebraicity and do yield the new determinant inequalities. They do not, by themselves, produce enough nodes or enough lower-bound divisibility to cross the four-exponentials threshold. The full conjecture should therefore remain marked open in this report.

2 Evidence boundary and repository provenance

2.1 Pinned state

The report is based on the repository state at

5d3982ceb12644fdd3499569d9db5f62f9a31dc6,

whose merge message records the reconciliation of the parallel conditional-structure and asymptotic formalizations. The source file specifically requested by the user is

Analysis/ExponentialIdentities/Article/alaoglu-erdos-combined-report/
alaoglu-erdos-combined-report.tex.

The authoritative formal import surface is `ExponentialIdentities.lean`.

This report inspected the cited Lean sources and theorem statements. It did not rerun the entire Lean build locally. Consequently, [\[repository kernel-verified Lean\]](#) means that the proof is present on the pinned repository import surface and is reported there as accepted by the Lean kernel; it is not a claim that a separate fresh build was performed while preparing this document.

2.2 Status labels

[repository kernel-verified Lean] A theorem is implemented in the pinned Lean corpus and appears on the aggregate import surface. The exact identifier is supplied whenever it is central.

[paper proof in this report] A conventional mathematical proof is given in this report. It is written with formalization in mind but has not yet been translated into Lean.

[interval computation] A finite interval computation described in the combined report. It has a stated software trust boundary and is not a kernel certificate.

[classical/open background] A standard theorem, an established open conjecture, or bibliographic background.

2.3 Current formal frontier

Result	Status	Principal Lean identifier or source
Three-base theorem $2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	[repository kernel-verified Lean]	<code>IntegerExponent.lean</code> ; interpolation determinant
Every nonintegral solution is transcendental	[repository kernel-verified Lean]	<code>transcendental_of_not_integer_of_two_three_rpow_integer</code>
$\vartheta = \log 3 / \log 2$ is transcendental	[repository kernel-verified Lean]	<code>transcendental_logThreeDivLogTwo</code>
No nonintegral solution with $2^x < 4096$	[repository kernel-verified Lean]	<code>FiniteCheck4096.lean</code>
Independent exclusion with $2^x < 2^{20}$	[interval computation]	Combined report interval scan; not a Lean proof
Four-exponentials implication	[repository kernel-verified Lean]	<code>alaogluErdosConjecture_of_realFourExponentialsConjecture</code>
Canonical primitive odd core and exact least generator	[repository kernel-verified Lean]	<code>exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture</code>
Exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$	[repository kernel-verified Lean]	Same primitive-generator theorem
Exact candidate monoid $\{2^n M^k\}$	[repository kernel-verified Lean]	Same primitive-generator theorem
Exact count in $[2^T, 2^{T+1})$	[repository kernel-verified Lean]	<code>twoBaseNaturalCandidatesInDyadicBlock_card</code>

Result	Status	Principal Lean identifier or source
Sharp quadratic cumulative asymptotic	[repository kernel-verified Lean]	tendsto_twoBaseNaturalCandidatesBelow_two_pow_div_sq
Growth equivalence for the conjecture	[repository kernel-verified Lean]	alaogluErdosConjecture_iff_candidate_count_below_two_pow_sub_quadratic
Vanishing block Fourier modes	[repository kernel-verified Lean]	exists_primitiveGenerator_with_vanishing_block_fourier_modes
Localized radical equivalence	[repository kernel-verified Lean]	alaogluErdosConjecture_iff_odd_rpow_not_localized
Primitive localized radical equivalence	[repository kernel-verified Lean]	alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion
Exact binomial minimal polynomial and radical degree	[repository kernel-verified Lean]	oddCoreRpow_exact_radical_degree
Simultaneous primitive output normalization	[repository kernel-verified Lean]	exists_simultaneous_primitive_output_normalization
Arbitrary-order arithmetic-progression obstruction	[repository kernel-verified Lean]	not_arithmetic_progression_twoBaseNaturalCandidates_of_curve
Arbitrary-node Vandermonde repulsion	[paper proof in this report]	Theorem 7.1
Semigroup generalized-Vandermonde hierarchy	[paper proof in this report]	Theorems 8.2 and 8.4
Explicit dyadic $O((\log X)^2)$ bound by that hierarchy	[paper proof in this report]	Theorem 8.8
Full Alaoglu–Erdős conjecture	Open	Equivalent radical and four-exponentials barriers below

2.4 Historical placement

Alaoglu and Erdős isolated the two-prime assertion in their 1944 study of highly composite and related numbers. They regarded it as very likely but could not prove it, and reported a communicated three-prime result of Siegel. In modern language, the three-base statement is supplied by the six exponentials theorem, whereas the two-base statement sits at the unresolved 2×2 threshold of the four exponentials conjecture [1, 8, 9]. Thus the elementary-looking integrality question is historically one of the canonical concrete tests of the gap between six and four exponentials, not an isolated recreational puzzle.

3 The problem in its most useful forms

3.1 Exponent form and candidate form

Conjecture 3.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Because 2^x and 3^x are positive, “integer” can be replaced by “positive natural number.” A solution cannot have $x < 0$, since then $0 < 2^x < 1$. Put

$$\vartheta := \frac{\log 3}{\log 2} = 1.584962500721156 \dots$$

and define the natural candidate set

$$\mathcal{C} := \{m \in \mathbb{N} : m > 0, m^\vartheta \in \mathbb{N}\}.$$

If $m = 2^x$, then

$$m^\vartheta = (2^x)^{\log 3 / \log 2} = 3^x.$$

Conversely, for $m > 0$, the exponent $x = \log_2 m$ satisfies $2^x = m$ and $3^x = m^\vartheta$. Therefore

$$\boxed{\text{Conjecture 3.1} \iff \mathcal{C} = \{2^n : n \in \mathbb{N}_0\}.} \tag{1}$$

This is the form used throughout the new determinant arguments.

3.2 The logarithmic determinant

Writing

$$M = 2^x, \quad A = 3^x,$$

a solution is equivalently a positive integer pair satisfying

$$\log M \log 3 - \log A \log 2 = 0. \tag{2}$$

The conjecture says that all positive integer solutions of (2) are

$$(M, A) = (2^n, 3^n), \quad n \in \mathbb{N}_0.$$

Thus the problem is a vanishing 2×2 determinant of logarithms of algebraic numbers. This is not merely an analogy with four exponentials; it is its exact local geometry.

3.3 One-parameter subgroup form

Let

$$\Gamma = \{(2^t, 3^t) : t \in \mathbb{R}\} \subset \mathbb{R}_{>0}^2.$$

Then the conjecture is

$$\Gamma \cap \mathbb{N}^2 = \{(2^n, 3^n) : n \in \mathbb{N}_0\}.$$

The curve is a real one-parameter subgroup of the multiplicative torus $\mathbb{G}_m^2(\mathbb{R})$. The difficulty is that its direction vector $(\log 2, \log 3)$ is transcendental rather than algebraic, so standard algebraic-subgroup intersection theorems do not apply directly.

3.4 Basic unconditional restrictions

The pinned corpus proves the following.

- A nonintegral x with $2^x \in \mathbb{Z}$ is irrational, and in fact transcendental by Gelfond–Schneider. Hence every counterexample is transcendental. [\[repository kernel-verified Lean\]](#)
- The number ϑ is transcendental. [\[repository kernel-verified Lean\]](#)
- Every counterexample satisfies $x \geq 12$ by the kernel-checked 4096 barrier. The separate interval computation in the combined report raises the checked barrier to $x > 20$ under its software trust assumptions. [\[repository kernel-verified Lean\]](#) [\[interval computation\]](#)
- The real four exponentials conjecture implies Conjecture 3.1. [\[repository kernel-verified Lean\]](#)

These facts eliminate rational, algebraic-irrational, small, and routine three-exponentials cases. They also explain why an elementary manipulation of logarithms is unlikely to finish the proof: the unknown lies exactly where present transcendence theory changes from theorem to conjecture.

4 What failure would look like: the exact conditional normal form

The most important fact in the current repository is not another exclusion range. It is the complete description of all solutions if even one nonintegral solution exists.

4.1 Canonical primitive generator

Let

$$\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{N}\}.$$

The theorem `exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture` proves the following conditional statement.

Theorem 4.1 (Exact conditional solution monoid). *Assume Conjecture 3.1 is false. Then there is a unique least nonintegral solution $\beta > 0$ such that*

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}, \tag{3}$$

and the representation is unique. If

$$M = 2^\beta, \quad A = 3^\beta,$$

then

$$\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}, \tag{4}$$

and this representation is also unique.

Status: [\[repository kernel-verified Lean\]](#)

This is a direct paraphrase of the unique-coordinate clauses in the cited primitive-generator theorem. The theorem also supplies the canonical odd core and radical normalization described next.

The phrase “least nonintegral” is stronger than choosing an arbitrary counterexample. It forces affine primitivity: no nontrivial common root can be extracted from the two output integers while preserving the same base-adic offset. This is the arithmetic analogue of choosing a primitive lattice direction.

4.2 Odd core, rational-power index, and localized radical

The same theorem supplies natural numbers w, d, a, c satisfying

$$\begin{aligned} w > 1, \quad w \text{ odd}, \quad d > 0, \quad c > 0, \\ \beta &= a + d \log_2 w, \\ 2^\beta &= 2^a w^d, \quad 3^\beta = c, \end{aligned}$$

with a gcd-normalization on the odd prime valuations of w . Define

$$E := w^\vartheta.$$

Then

$$E^d = \frac{c}{3^a}, \quad a = 0 \quad \text{or} \quad 3 \nmid c, \tag{5}$$

and d is the least positive exponent for which E^d is rational. The module `RadicalDegree.lean` proves

$$\text{minpoly}_{\mathbb{Q}}(E) = X^d - \frac{c}{3^a}, \quad [\mathbb{Q}(E) : \mathbb{Q}] = d, \tag{6}$$

with irreducibility of the displayed binomial.

Consequently the conjecture is kernel-verified equivalent to each of the following exclusion statements.

1. For every odd $u > 1$,

$$u^\vartheta \notin \mathbb{Z}[1/3]. \tag{7}$$

2. It is enough to quantify over odd $u > 1$ that are not proper powers.

3. There are no power-primitive odd $w > 1$, $d > 0$, and $a, B \in \mathbb{N}_0$ such that

$$3^a (w^\vartheta)^d = B. \tag{8}$$

This is the cleanest remaining target. It is only one real number at a time, but already the special choice $u = 3$ asks whether $3^{\log_2 3}$ can lie in $\mathbb{Z}[1/3]$, which is not decided by current transcendence theorems.

4.3 Simultaneous primitive normalization of both outputs

The module `OutputNormalization.lean` gives a symmetric refinement. For the least β , one may write

$$M = 2^a w^d, \quad A = 3^b v^e,$$

where w is odd and power-primitive, $3 \nmid v$, v is power-primitive, and

$$a = 0 \quad \text{or} \quad b = 0. \tag{9}$$

Moreover,

$$\gcd(d, e, |b - a|) = 1. \quad (10)$$

The outer exponents d and e are intrinsic: they are recovered as gcds of the prime valuations of the base-coprime parts of M and A .

These restrictions are strong, but they still admit infinitely many formal integer parameter patterns. The unresolved analytic equation remains

$$\frac{\log M}{\log 2} = \frac{\log A}{\log 3}.$$

No known local valuation argument turns (9) and (10) into a contradiction.

4.4 Exact counts and the conditional population

The monoid (3) makes the population of a hypothetical counterexample completely explicit. In every unit exponent block $[T, T + 1)$,

$$\#(\mathcal{S} \cap [T, T + 1)) = 1 + \left\lfloor \frac{T + 1}{\beta} \right\rfloor. \quad (11)$$

Passing through $m = 2^x$, this becomes the exact dyadic magnitude count

$$\#(\mathcal{C} \cap [2^T, 2^{T+1})) = 1 + \left\lfloor \frac{T + 1}{\beta} \right\rfloor. \quad (12)$$

Thus a failed conjecture would produce only linearly many candidates in the logarithmic block index:

$$\#(\mathcal{C} \cap [X, 2X]) \asymp \log X.$$

Cumulatively,

$$\#(\mathcal{C} \cap [1, 2^T)) = \frac{T^2}{2\beta} + O(T). \quad (13)$$

The repository formalizes both the exact floor sums and the limiting constant $1/(2\beta)$. It also proves the growth reformulation

$$\text{Conjecture 3.1} \iff \frac{\#(\mathcal{C} \cap [1, 2^T))}{T^2} \longrightarrow 0. \quad (14)$$

This exact population is the correct benchmark for every proposed density or determinant argument. A method proving only $O((\log X)^2)$ candidates in $[X, 2X]$, for example, is a valid unconditional theorem but cannot contradict (12).

4.5 Blockwise irrational rotation

For fixed T , the nonintegral solutions in $[T, T + 1)$ have fractional parts drawn from the initial orbit segment

$$\{\beta\}, \{2\beta\}, \dots, \left\{ \left\lfloor \frac{T + 1}{\beta} \right\rfloor \beta \right\}.$$

Because β is irrational, the repository proves vanishing of every nonzero normalized Fourier mode on these blocks. The combined report derives the corresponding continuous-test and interval-counting equidistribution at paper level.

This attractive regularity does not solve the conjecture. The exact integrality condition is not a fixed positive-measure target in the circle; it is an exponentially shrinking target. That mismatch will recur below.

5 The exact four-exponentials barrier

5.1 The 2×2 exponential matrix

Suppose x is a nonintegral solution. Consider row vectors $(1, x)$ and column vectors $(\log 2, \log 3)$. The four corresponding exponentials are

$$\exp(\log 2) = 2, \quad \exp(\log 3) = 3, \quad \exp(x \log 2) = 2^x, \quad \exp(x \log 3) = 3^x.$$

All four are algebraic, indeed positive integers. The rows are linearly independent over $\mathbb{Q}$ because x is irrational; the columns are linearly independent because no nonzero powers of 2 and 3 agree. The real four exponentials conjecture predicts that at least one of the four exponentials must be transcendental, a contradiction [8, 9].

The Lean theorem `alaogluErdosConjecture_of_realFourExponentialsConjecture` formalizes exactly this implication. It does not assume four exponentials as an axiom; it defines the conjectural proposition and proves the conditional implication.

5.2 Why six exponentials does not specialize

The six exponentials theorem applies to either two independent row parameters and three independent column parameters, or three rows and two columns. Here only two canonical logarithmic directions are available. Adding a third base 5 gives the theorem

$$2^x, 3^x, 5^x \in \mathbb{Z} \implies x \in \mathbb{Z},$$

which is already formalized by an interpolation determinant. In the two-base problem, any additional base built from powers of 2 and 3 has logarithm in their rational span and does not create the missing independent column.

The module `ThreeSmooth.lean` and the denominator-controlled results in `RationalThirdBase.lean` make this obstruction precise: an auxiliary base with a prime factor outside $\{2, 3\}$ would be powerful, but its corresponding exponential value is not forced by the original hypotheses. Bases supported only on 2 and 3 do not increase the transcendence rank.

5.3 The integral special case may still be easier — but not automatically

Four exponentials asks only for algebraicity. The present values are integers, so there is extra discrete structure. The determinant results in [sections 7](#) and [8](#) extract some of that structure: nonzero determinants are integers and therefore have absolute value at least 1. However, the lower

bound 1 is the only universally available archimedean information. To beat the four-exponentials threshold, one likely needs to show that the determinants are divisible by a large integer, or that a suitable common denominator is anomalously small. That observation motivates the p -adic research priority in [section 11](#).

6 Full-proof attempts and their failure points

This section records the principal avenues explored before turning to the new determinant results. The purpose is not merely negative. Each failure point identifies a quantitative or structural feature that a successful proof must improve.

6.1 Rational approximation to a least generator

Assume failure and let β be the least nonintegral solution, with $M = 2^\beta$ and $A = 3^\beta$. Let p/q approximate β , and put $\varepsilon = q\beta - p$. Then

$$M^q - 2^p = 2^p \left(e^{\varepsilon \log 2} - 1 \right), \quad (15)$$

$$A^q - 3^p = 3^p \left(e^{\varepsilon \log 3} - 1 \right). \quad (16)$$

Both left sides are integers. If either is nonzero, its absolute value is at least 1. For small ε , this gives bounds of the shape

$$|\varepsilon| \gtrsim 2^{-p} \quad \text{and} \quad |\varepsilon| \gtrsim 3^{-p}.$$

These are exponentially tiny in q . Continued fractions only guarantee infinitely many approximants with $|\varepsilon| \ll q^{-1}$, while Baker-type lower bounds for ordinary linear forms in logarithms are polynomial in the relevant heights [2, 8]. Neither scale conflicts with an exponential lower bound as weak as M^{-q} or A^{-q} .

One might hope to multiply (15) and (16), or eliminate ε between them. To first order their ratio merely recovers $\log 3 / \log 2$; higher terms are too small to create an integer contradiction. The two integer gaps are correlated, not independent.

Observation 6.1. Any rational-approximation proof needs more than “a nonzero integer has size at least one.” It needs a divisibility lower bound growing exponentially with q , or an approximation to β exponentially better than continued fractions can supply in general.

6.2 Equidistribution versus exponentially shrinking targets

The multiples $\{k\beta\}$ are equidistributed modulo 1. Meanwhile, integrality of 2^{n+t} between adjacent integral powers forces the fractional part t away from both endpoints by an amount comparable with 2^{-n} ; the base-3 condition gives a comparable 3^{-n} restriction. For the k th multiple, the relevant target width is therefore exponentially small in k .

Equidistribution controls visits to fixed intervals and, with discrepancy estimates, to moderately shrinking intervals. It does not force visits to a summable sequence of targets whose lengths behave like e^{-ck} . Indeed, the Borel–Cantelli heuristic predicts only finitely many such visits for a typical rotation. The exact conditional monoid is perfectly compatible with this behavior.

The repository’s blockwise Weyl theorems are therefore structurally correct but point in the wrong direction for a contradiction: they describe the distribution of already existing solutions within each block, not the probability that arbitrary exponents hit the integer lattice.

6.3 Finite differences

For $f(t) = t^\vartheta$, the repository proves the exact mean-value formula for every forward difference order r :

$$\Delta_h^r f(n) = (\vartheta)_{\underline{r}} h^r \xi^{\vartheta-r} \quad (n < \xi < n + rh).$$

If all $r + 1$ values $f(n), f(n + h), \dots, f(n + rh)$ are integers, then the left side is an integer. Under

$$|(\vartheta)_{\underline{r}}| h^r < n^{r-\vartheta},$$

its absolute value is strictly between 0 and 1, a contradiction. This is a sharp and uniform obstruction to candidate arithmetic progressions.

The limitation is combinatorial. A rank-two multiplicative monoid need not contain long additive arithmetic progressions among its values. Even many points in a short interval need not be equally spaced. The new arbitrary-node determinant in [section 7](#) removes that equal-spacing hypothesis, but its packing exponent remains too weak to contradict the exact conditional count.

6.4 Radical fields and conjugates

The normalized radical $E = w^\vartheta$ is algebraic and has exact minimal polynomial $X^d - c/3^a$. Its conjugates are $\zeta_d^j E$. This gives complete algebraic control of the field $\mathbb{Q}(E)$, its degree, and its norm. What it does not give is a way to transport the real analytic identity

$$\log E = \vartheta \log w$$

to the other conjugates. The chosen positive real logarithm is not Galois-equivariant; the other roots differ by complex logarithmic periods. Taking norms merely reproduces $E^d = c/3^a$.

Ramification is similarly inconclusive. The primes occurring in the binomial and in w are highly constrained, but the equation relates them through a quotient of logarithms, not through an algebraic field embedding. Standard Kummer theory sees the radical equation but not the special value $\vartheta = \log 3 / \log 2$.

Observation 6.2. A radical-field proof needs a genuinely mixed invariant: something simultaneously sensitive to the algebraic conjugates of E and to the logarithmic relation among $2, 3, w, E$. Separate algebraic and real-analytic arguments currently meet only at the four-logarithm determinant.

6.5 Compact S -integral orbit

From the least generator define

$$U_k := \frac{M^k}{2^{[k\beta]}} = 2^{\{k\beta\}} \in \mathbb{Z}[1/2], \quad V_k := \frac{A^k}{3^{[k\beta]}} = 3^{\{k\beta\}} \in \mathbb{Z}[1/3].$$

The points (U_k, V_k) lie on the compact analytic arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}$$

and are dense there because β is irrational. Their logarithmic heights grow linearly in k , while the number of points of logarithmic height at most H is also linear in H .

This suggests o-minimal and S -integral counting. Butler’s work relates special cases of Wilkie-type algebraic-point counting to real three- and four-exponentials statements [3]. The standard Pila–Wilkie conclusion, however, is subpolynomial in the usual multiplicative height outside algebraic parts. Reparameterized by logarithmic height, the orbit’s linear count is not immediately contradictory, especially because the denominators are highly structured and grow exponentially.

A promising refinement would count points on $\mathcal{A}$ with denominator supports exactly $\{2\}$ and $\{3\}$ and with synchronized denominator exponents. No theorem of the required strength is currently available in the inspected corpus.

6.6 Forcing an auxiliary base

If one could exhibit an integer $b > 1$ multiplicatively independent of 2 and 3 such that b^x is algebraic (or rational with suitably controlled denominator), then six exponentials or the repository’s rational-third-base theorems could force $x \in \mathbb{Q}$, hence $x \in \mathbb{Z}$. The original hypotheses provide no such b .

The obvious constructions $b = 2^u 3^v$ remain in the same logarithmic span. Constructions from $M = 2^x$ and $A = 3^x$ also collapse to that span after taking logarithms. A successful auxiliary base must therefore arise from a nontrivial arithmetic operation whose exponential behavior is controlled but whose logarithm leaves the rank-two span. This is essentially an algebraic independence problem in disguise.

7 New result I: arbitrary-node lattice repulsion

The repository’s higher-difference theorem uses equally spaced nodes. The same integer-versus-small-real principle extends to arbitrary nodes through interpolation determinants.

7.1 The determinant identity

For distinct real numbers $x_0 < \dots < x_r$ and a function f , write $f[x_0, \dots, x_r]$ for the divided difference. The standard determinant identity is

$$\det \begin{pmatrix} 1 & x_0 & \dots & x_0^{r-1} & f(x_0) \\ 1 & x_1 & \dots & x_1^{r-1} & f(x_1) \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & x_r & \dots & x_r^{r-1} & f(x_r) \end{pmatrix} = V(x_0, \dots, x_r) f[x_0, \dots, x_r], \quad (17)$$

where

$$V(x_0, \dots, x_r) = \prod_{0 \leq i < j \leq r} (x_j - x_i).$$

If $f \in C^r$ on the interval spanned by the nodes, the generalized mean-value theorem gives some $\xi \in (x_0, x_r)$ such that

$$f[x_0, \dots, x_r] = \frac{f^{(r)}(\xi)}{r!}. \quad (18)$$

For $f(t) = t^\vartheta$,

$$f^{(r)}(t) = (\vartheta)_{\underline{r}} t^{\vartheta-r}, \quad (\vartheta)_{\underline{r}} := \prod_{j=0}^{r-1} (\vartheta - j).$$

Since ϑ is not an integer, the coefficient never vanishes.

7.2 Arbitrary-node theorem

Theorem 7.1 (Vandermonde repulsion for candidates). *Let $r \geq 2$ and let*

$$1 \leq m_0 < m_1 < \dots < m_r$$

be natural candidates, so $m_i^\vartheta \in \mathbb{N}$. Then

$$\prod_{0 \leq i < j \leq r} (m_j - m_i) \geq \frac{r!}{|(\vartheta)_{\underline{r}}|} m_0^{r-\vartheta}. \quad (19)$$

Proof. Apply (17) to $f(t) = t^\vartheta$ and the nodes m_i . Every entry of the matrix is an integer: the ordinary powers are integral by construction and the last column is integral because the m_i are candidates. By (18), its determinant is

$$D = V(m_0, \dots, m_r) \frac{(\vartheta)_{\underline{r}}}{r!} \xi^{\vartheta-r} \quad (m_0 < \xi < m_r).$$

It is nonzero, so $D \in \mathbb{Z} \setminus \{0\}$ and $|D| \geq 1$. Because $r \geq 2 > \vartheta$, the exponent $\vartheta - r$ is negative and

$$\xi^{\vartheta-r} \leq m_0^{\vartheta-r}.$$

Therefore

$$1 \leq |D| \leq V(m_0, \dots, m_r) \frac{|(\vartheta)_{\underline{r}}|}{r!} m_0^{\vartheta-r},$$

which rearranges to (19). □

Status: [paper proof in this report]

The argument is unconditional and elementary once the divided-difference mean-value theorem is available. It is not present in the inspected Lean modules; `HigherDifference.lean` uses arithmetic progressions. A direct formalization blueprint appears in [section 12](#).

7.3 Three points: a lattice-triangle inequality

For $r = 2$, the determinant is

$$D = \det \begin{pmatrix} 1 & m_0 & m_0^\vartheta \\ 1 & m_1 & m_1^\vartheta \\ 1 & m_2 & m_2^\vartheta \end{pmatrix}.$$

Geometrically, $|D|$ is twice the area of the lattice triangle with vertices (m_i, m_i^ϑ) . Strict convexity makes the area positive; lattice integrality makes it at least $1/2$.

Corollary 7.2 (Three-candidate span). *If three candidates lie in $[N, N + H]$, where $N \geq 1$, then*

$$H \geq \left(\frac{8}{\vartheta(\vartheta - 1)} \right)^{1/3} N^{(2-\vartheta)/3}. \quad (20)$$

Numerically,

$$H \geq 2.0510723956 \dots N^{0.1383458330 \dots}.$$

Proof. Write $a = m_1 - m_0$, $b = m_2 - m_1$. Then $a, b > 0$, $a + b \leq H$, and

$$V = ab(a + b) \leq \frac{H^3}{4},$$

with the continuous maximum at $a = b = H/2$. Theorem 7.1 at $r = 2$ gives

$$V \geq \frac{2}{\vartheta(\vartheta - 1)} N^{2-\vartheta}.$$

Combining the inequalities yields (20). □

7.4 General local packing bound

Put

$$R_r := \binom{r+1}{2} = \frac{r(r+1)}{2}, \quad C_r := \left(\frac{r!}{|(\vartheta)_r|} \right)^{1/R_r}, \quad \alpha_r := \frac{r - \vartheta}{R_r}.$$

Since every pairwise gap among points in $[N, N + H]$ is at most H , the Vandermonde product is at most H^{R_r} . Theorem 7.1 immediately gives:

Corollary 7.3 (At most r points below the repulsion scale). *If $r + 1$ candidates lie in $[N, N + H]$, then*

$$H \geq C_r N^{\alpha_r}. \quad (21)$$

Equivalently, every interval of length strictly less than $C_r N^{\alpha_r}$ beginning at or after N contains at most r candidates.

The exponent can be optimized. As a function of a real variable $r > \vartheta$,

$$\alpha(r) = \frac{2(r - \vartheta)}{r(r + 1)}$$

has its maximum at

$$r = \vartheta + \sqrt{\vartheta^2 + \vartheta} = 3.60908 \dots$$

Hence the best integer order in this ordinary-power family is $r = 4$.

In particular:

r	R_r	$ (\vartheta)_r $	α_r	C_r
2	3	0.9271436280	0.1383458331	1.2920946431
3	6	0.3847993728	0.2358395832	1.5805909191
4	10	0.5445055422	0.2415037499	1.4602287461
5	15	1.3150013031	0.2276691666	1.3510881062
6	21	4.4907787617	0.2102398809	1.2735045942
7	28	19.8269566337	0.1933941964	1.2187063909
8	36	107.3637136680	0.1781954861	1.1790125192

Table 2: Ordinary-power arbitrary-node repulsion constants. For $r = 2$, the sharper three-point constant in [theorem 7.2](#) uses the exact maximum of the three-gap product.

Corollary 7.4 (Best ordinary-power local statement). *Every interval*

$$[N, N + H], \quad H < 1.4602287460 N^{0.2415037499},$$

contains at most four natural candidates.

7.5 A bound for any longer interval

The span estimate can be converted into a counting inequality. Let $m_1 < \dots < m_s$ be all candidates in $[N, N + H]$. For each $i \leq s - r$, the consecutive window $m_i, \dots, m_{i+r}$ has span at least $C_r N^{\alpha_r}$. Summing these spans, each adjacent gap $m_{j+1} - m_j$ is counted at most r times. Hence

$$(s - r)C_r N^{\alpha_r} \leq r(m_s - m_1) \leq rH.$$

Corollary 7.5 (Arbitrary-node counting inequality). *For every $r \geq 2$,*

$$\#(\mathcal{C} \cap [N, N + H]) \leq r + \frac{r}{C_r} H N^{-\alpha_r}. \quad (22)$$

This is a genuine strengthening of progression avoidance: it constrains every possible configuration. It is still far from the conditional benchmark. At the optimal $r = 4$ and $H = N$, (22) is $O(N^{0.7585\dots})$, whereas the exact failed-conjecture population would be only $O(\log N)$.

7.6 Relation to higher differences

For equally spaced nodes $m_i = n + ih$, the Vandermonde factor is explicit:

$$V = h^{R_r} \prod_{j=1}^r j!.$$

Substitution in (17) recovers an integer interpolation remainder closely related to the r th forward difference. The repository theorem is sharper for this special configuration because it uses the exact binomial combination rather than bounding all gaps by a common span. Conversely, Theorem 7.1 applies when the nodes have no additive structure at all. The two results are complementary.

8 New result II: semigroup generalized Vandermonde determinants

The preceding determinant used only the integral columns $1, m, m^2, \dots$ and one extra integral column m^ϑ . But a candidate carries much more integral data. If $m^\vartheta = n \in \mathbb{N}$, then for every $a, b \in \mathbb{N}_0$,

$$m^{a+b\vartheta} = m^a n^b \in \mathbb{N}. \quad (23)$$

The full set of available exponents is therefore

$$\Lambda_\vartheta := \{a + b\vartheta : a, b \in \mathbb{N}_0\}.$$

Since ϑ is irrational, all pairs (a, b) give distinct exponents.

8.1 Positivity of generalized Vandermonde determinants

Let

$$0 \leq \lambda_0 < \lambda_1 < \dots < \lambda_r \quad \text{and} \quad 0 < x_0 < x_1 < \dots < x_r.$$

The functions x^{λ_j} form an extended complete Chebyshev system on $(0, \infty)$. Their Wronskian is [4]

$$W(x) = x^{\sum_j \lambda_j - Rr} \prod_{0 \leq i < j \leq r} (\lambda_j - \lambda_i) > 0. \quad (24)$$

Indeed, after factoring x^{λ_j} from column j and x^{-i} from derivative row i , the remaining determinant is the Vandermonde determinant of the λ_j , expressed in the falling-factorial basis. Positivity of all initial Wronskians implies strict positivity of every ordered evaluation determinant.

For completeness, there is also an elementary zero-count proof. Put $x = e^t$. A nonzero linear combination of x^{λ_j} becomes an exponential polynomial $\sum c_j e^{\lambda_j t}$. After factoring the lowest exponential, differentiation removes the constant term and leaves one fewer exponential. Induction with Rolle's theorem shows that a combination of $r + 1$ such functions has at most r real zeros counting multiplicity. Consequently an evaluation determinant at $r + 1$ ordered nodes cannot vanish. Its sign is constant on the connected chamber $x_0 < \dots < x_r$ and is positive in the coalescing-node limit by (24).

Lemma 8.1 (Generalized Vandermonde positivity). *For strictly increasing nonnegative exponents λ_j and strictly increasing positive nodes x_i ,*

$$\det(x_i^{\lambda_j})_{0 \leq i, j \leq r} > 0. \quad (25)$$

8.2 Integer determinants from candidates

Theorem 8.2 (Semigroup integer determinant). *Let $m_0 < \dots < m_r$ be natural candidates. Choose distinct pairs $(a_j, b_j) \in \mathbb{N}_0^2$ and order*

$$\lambda_j = a_j + b_j \vartheta$$

so that $\lambda_0 < \dots < \lambda_r$. Then

$$D_\lambda(\mathbf{m}) := \det(m_i^{\lambda_j})_{0 \leq i, j \leq r} \quad (26)$$

is a positive integer. In particular, $D_\lambda(\mathbf{m}) \geq 1$.

Proof. By (23), each entry equals

$$m_i^{a_j} (m_i^{\vartheta})^{b_j} \in \mathbb{N},$$

so the determinant is an integer. It is strictly positive by Lemma 8.1. $\square$

Status: [\[paper proof in this report\]](#)

This theorem packages all mixed integrality consequences into one reusable object. It is not the same as the repository’s exponential interpolation determinant in `MultiplicativeRank.lean`: here the rows are arbitrary natural candidates and the columns are chosen from the two-dimensional exponent semigroup.

8.3 Divided-difference factorization

Let $f_j(x) = x^{\lambda_j}$ and let

$$V(\mathbf{m}) = \prod_{i < j} (m_j - m_i).$$

Applying the Newton divided-difference row transformation gives

$$D_{\lambda}(\mathbf{m}) = V(\mathbf{m}) \det(f_j[m_0, \dots, m_i])_{0 \leq i, j \leq r}. \quad (27)$$

The transformation is exact; its determinant is $V(\mathbf{m})^{-1}$.

Suppose

$$N \leq m_0 < \dots < m_r \leq N + H \leq 2N.$$

For each divided difference, the mean-value theorem gives a point in $[N, 2N]$ and

$$|f_j[m_0, \dots, m_i]| \leq \frac{|(\lambda_j)_i|}{i!} \sup_{N \leq x \leq 2N} x^{\lambda_j - i}. \quad (28)$$

We need a uniform elementary coefficient estimate.

Lemma 8.3 (Falling-factorial coefficient bound). *For $\lambda \geq 0$ and $i \in \mathbb{N}_0$,*

$$\frac{|(\lambda)_i|}{i!} \leq 2^{\lambda+i}. \quad (29)$$

Proof. If $0 \leq \lambda \leq 1$, each factor satisfies $|\lambda - k| \leq k + 1$, so the quotient is at most 1. If $\lambda \geq 1$, let $n = \lceil \lambda \rceil$. Then

$$\frac{|(\lambda)_i|}{i!} \leq \prod_{k=0}^{i-1} \frac{\lambda + k}{k + 1} \leq \prod_{k=0}^{i-1} \frac{n + k}{k + 1} = \binom{n + i - 1}{i}.$$

The last binomial coefficient is at most the sum of its row, 2^{n+i-1} , and $n - 1 \leq \lambda$. Hence it is at most $2^{\lambda+i}$. $\square$

For $x \in [N, 2N]$,

$$x^{\lambda-i} \leq 2^{\lambda} N^{\lambda-i}. \quad (30)$$

If $\lambda - i \geq 0$, use $x \leq 2N$; if it is negative, use $x \geq N$. Combining (28), (29), and (30),

$$|f_j[m_0, \dots, m_i]| \leq 2^{2\lambda_j+i} N^{\lambda_j-i}. \quad (31)$$

8.4 Quantitative semigroup gap theorem

Set

$$R := R_r = \frac{r(r+1)}{2}, \quad \Sigma := \sum_{j=0}^r \lambda_j.$$

By the Leibniz formula and (31), every permutation term in the small determinant in (27) is at most

$$2^{2\Sigma+R} N^{\Sigma-R},$$

and there are $(r+1)!$ terms. Since every pairwise node difference is at most H , $V(\mathbf{m}) \leq H^R$. The positive integer lower bound from Theorem 8.2 now gives the following.

Theorem 8.4 (Semigroup determinant gap). *Let $r \geq 1$. Let $\lambda_0 < \dots < \lambda_r$ be any $r+1$ exponents in Λ_ϑ , and put $R = r(r+1)/2$ and $\Sigma = \sum_j \lambda_j$. If $r+1$ natural candidates lie in*

$$[N, N+H] \subseteq [N, 2N], \quad N \geq 1,$$

then

$$H \geq ((r+1)!)^{-1/R} 2^{-1-2\Sigma/R} N^{1-\Sigma/R}. \quad (32)$$

Proof. Equations (27) and (31) imply

$$1 \leq D_\lambda(\mathbf{m}) \leq H^R (r+1)! 2^{R+2\Sigma} N^{\Sigma-R}.$$

Taking R th roots and rearranging proves (32). $\square$

The exponent $1 - \Sigma/R$ is the key. We should choose the $r+1$ smallest available semigroup exponents to minimize Σ .

8.5 The ordered exponent semigroup

Write the elements of Λ_ϑ in increasing order:

$$0 = \lambda_0 < \lambda_1 < \lambda_2 < \dots$$

The beginning is

$$0, 1, \vartheta, 2, 1+\vartheta, 3, 2\vartheta, 2+\vartheta, 4, 1+2\vartheta, 3+\vartheta, 3\vartheta, 5, \dots$$

The counting function is elementary:

$$\begin{aligned} L(T) &:= \#\{\lambda \in \Lambda_\vartheta : \lambda \leq T\} \\ &= \sum_{0 \leq b \leq T/\vartheta} (|T - b\vartheta| + 1) = \frac{T^2}{2\vartheta} + O(T). \end{aligned} \quad (33)$$

Irrationality of ϑ ensures that different pairs (a, b) are counted distinctly. Inverting (33) and summing the order statistics gives

$$\lambda_r = \sqrt{2\vartheta r} + O(1), \quad (34)$$

$$\Sigma_r := \sum_{j=0}^r \lambda_j = \frac{2}{3} \sqrt{2\vartheta} r^{3/2} + O(r), \quad (35)$$

$$\frac{\Sigma_r}{R_r} = \frac{4}{3} \sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \quad (36)$$

Consequently the power of N in (32) tends to 1:

$$1 - \frac{\Sigma_r}{R_r} = 1 - \frac{4}{3} \sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \quad (37)$$

r	Σ_r	$1 - \Sigma_r/R_r$
2	2.5849625007	0.1383458331
3	4.5849625007	0.2358395832
4	7.1699250014	0.2830074999
5	10.1699250014	0.3220049999
6	13.3398500029	0.3647690475
7	16.9248125036	0.3955424106
8	20.9248125036	0.4187552082
9	25.0947375050	0.4423391666
10	29.6797000058	0.4603690908
12	39.4345875079	0.4944283653
20	87.7939875195	0.5819333928

Table 3: Exponent gain from using the first $r + 1$ mixed exponents in $\mathbb{N}_0 + \mathbb{N}_0\vartheta$. The ordinary-power family peaks and then decays; the semigroup family improves monotonically in these data and tends to exponent 1.

8.6 An explicit coarse form

For the global counting corollary we need no delicate lattice-point error term. A simple bound suffices.

Lemma 8.5 (Crude semigroup weight bound). *For $r \geq 2$, the first $r + 1$ semigroup exponents satisfy*

$$\lambda_r \leq 6\sqrt{r}, \quad \frac{\Sigma_r}{R_r} \leq \frac{12}{\sqrt{r}}. \quad (38)$$

Proof. Let $q = \lceil \sqrt{r+1} \rceil$. The $(q+1)^2$ pairs $0 \leq a, b \leq q$ give distinct exponents and each is at most $q(1+\vartheta) < 3q$. Since $(q+1)^2 > r+1$, the r th ordered exponent is below $3q$. For $r \geq 2$, $q \leq \sqrt{r+1} + 1 \leq 2\sqrt{r}$, hence $\lambda_r \leq 6\sqrt{r}$. Then

$$\Sigma_r \leq (r+1)\lambda_r \leq 6(r+1)\sqrt{r},$$

and division by $R_r = r(r+1)/2$ gives the second inequality. $\square$

For $r \geq 144$, (38) gives $\Sigma_r/R_r \leq 1$. Also

$$((r+1)!)^{1/R_r} \leq (r+1)^{(r+1)/R_r} = (r+1)^{2/r} \leq 3$$

for $r \geq 2$. Therefore Theorem 8.4 yields a particularly clean form.

Corollary 8.6 (Explicit near-linear span). *If $r \geq 144$ and $r+1$ natural candidates lie in $[N, N+H] \subseteq [N, 2N]$, then*

$$H \geq \frac{1}{24} N^{1-12/\sqrt{r}}. \quad (39)$$

Proof. In (32), use $((r+1)!)^{-1/R} \geq 1/3$, $2^{-1-2\Sigma/R} \geq 2^{-3} = 1/8$, and, since $N \geq 1$, $N^{1-\Sigma/R} \geq N^{1-12/\sqrt{r}}$. $\square$

8.7 Counting candidates in a general interval

As in theorem 7.5, sum the spans of consecutive windows of $r+1$ candidates. Every adjacent gap is counted at most r times.

Corollary 8.7 (Semigroup interval count). *For $r \geq 144$ and $1 \leq H \leq N$,*

$$\#(\mathcal{C} \cap [N, N+H]) \leq r + 24rHN^{-1+12/\sqrt{r}}. \quad (40)$$

Proof. If s is the number of candidates and $s \leq r$, there is nothing to prove. Otherwise, Corollary 8.6 gives a lower bound $L = N^{1-12/\sqrt{r}}/24$ for each span $m_{i+r} - m_i$. Thus

$$(s-r)L \leq rH,$$

which is equivalent to (40). $\square$

8.8 An independent dyadic logarithmic-square bound

Corollary 8.8 (Explicit $O((\log X)^2)$ dyadic bound). *For every real $X \geq 1$,*

$$\#(\mathcal{C} \cap [X, 2X]) \leq 10000 \max\{1, \log X\}^2. \quad (41)$$

Proof. Put $L = \max\{1, \log X\}$ and

$$r = \lceil 144L^2 \rceil.$$

Then $r \geq 144$, $\sqrt{r} \geq 12L$, and

$$X^{12/\sqrt{r}} = \exp\left(\frac{12 \log X}{\sqrt{r}}\right) \leq e.$$

Apply (40) with $N = H = X$:

$$\#(\mathcal{C} \cap [X, 2X]) \leq r + 24rX^{12/\sqrt{r}} \leq (1 + 24e)r.$$

Since $L \geq 1$, $r \leq 145L^2$, and $(1 + 24e)145 < 10000$. $\square$

Status: [paper proof in this report]

The bound is numerically coarse by design. Its value is conceptual independence: it comes from positivity and integrality of generalized Vandermonde determinants, not from the prime-factor valuation rank of candidates. Refining constants is much less important than finding an additional source of determinant lower bound.

9 Why the new hierarchy still stops short

The semigroup determinant is much stronger locally than the ordinary-power determinant: its repulsion exponent approaches 1. It is therefore important to identify exactly why it does not prove the conjecture.

9.1 How many nodes are needed to approach scale X ?

The asymptotic exponent in (37) is

$$\delta_r := 1 - \frac{\Sigma_r}{R_r} = 1 - \frac{c_\theta}{\sqrt{r}} + O(r^{-1}), \quad c_\theta = \frac{4}{3}\sqrt{2\vartheta} = 2.373\dots$$

For nodes near X , the lower span has the shape

$$X^{\delta_r} = X \exp\left(-\frac{c_\theta \log X}{\sqrt{r}} + O\left(\frac{\log X}{r}\right)\right).$$

To make the exponential loss in (9.1) bounded independently of X , one needs

$$\sqrt{r} \asymp \log X, \quad \text{that is,} \quad r \asymp (\log X)^2.$$

This is precisely why Corollary 8.8 has a logarithmic square.

9.2 How many nodes a counterexample supplies

Conditional on a counterexample, the exact block count (12) gives

$$\#(\mathcal{C} \cap [X, 2X]) = \frac{\log_2 X}{\beta} + O(1) \asymp \log X.$$

Thus the actual conditional population is smaller than the determinant threshold (9.1) by one factor of $\log X$.

Taking $r \asymp \log X$ in (9.1) yields only

$$X \exp(-C\sqrt{\log X}),$$

which is much smaller than the average spacing $X/\log X$ among the $\asymp \log X$ conditional candidates. The inequality is therefore compatible with the exact monoid even at the level of expected spacing.

9.3 Broader multiplicative intervals do not help

The cumulative conditional count below X is $\asymp (\log X)^2$, so one might try to use all those points at once. But their nodes range from 1 to X , and the gap theorem depends on the smallest node. Restricting to $[X^\gamma, X]$ with fixed $0 < \gamma < 1$ does retain $\asymp (\log X)^2$ points, but its additive length is $\asymp X$, while the lower scale is at most $X^{\gamma+o(1)}$. Again there is no contradiction.

This is not a mere defect of coarse constants. It reflects a mismatch between additive node geometry and the logarithmic triangular lattice of exponents (n, k) in $m = 2^n M^k$.

9.4 The determinant needs one more source of size

The archimedean argument uses only

$$D \in \mathbb{Z}_{>0} \implies D \geq 1.$$

If the exact monoid coordinates forced

$$D \equiv 0 \pmod{Q} \quad \text{with} \quad \log Q \gg R \log X / \sqrt{r},$$

then the lower bound $D \geq Q$ could remove the one-logarithm deficit. Alternatively, one could seek a determinant with total exponent weight $\Sigma = O(r)$ rather than $\Sigma \asymp r^{3/2}$, but the two-dimensional semigroup counting law $L(T) \asymp T^2$ makes that impossible using only distinct nonnegative mixed monomials.

This gives a concrete design specification:

A successful determinant proof must add nonarchimedean divisibility, introduce lower-weight functions outside the mixed-monomial semigroup, or exploit the exact two-dimensional row grid so efficiently that $O(\log X)$ nodes suffice.

10 Further equivalent reformulations

Several equivalent forms are already formalized in the repository. The exact conditional structure also yields useful new paper-level reformulations that expose different possible proof technologies.

10.1 Integer points on a transcendental lattice curve

The candidate formulation (1) is

$$\{(m, n) \in \mathbb{N}^2 : n = m^\vartheta\} = \{(2^k, 3^k) : k \in \mathbb{N}_0\}. \quad (42)$$

The curve is strictly convex and transcendental. Theorem 7.1 can be read as a lower bound for lattice simplex volumes cut from this curve.

10.2 Vanishing determinant of four logarithms

As in (2), the conjecture is equivalent to

$$\log M \log 3 - \log A \log 2 = 0, \quad M, A \in \mathbb{N}, \quad \implies \quad (M, A) = (2^k, 3^k). \quad (43)$$

This is the most direct bridge to four exponentials and algebraic independence of logarithms.

10.3 Primitive localized-radical exclusion

The kernel-verified radical form is

$$\forall w > 1 \text{ odd and power-primitive}, \quad \forall d > 0, \quad a \geq 0, \quad 3^a(w^\vartheta)^d \notin \mathbb{N}. \quad (44)$$

If this fails with least rational-power index d , then $X^d - c/3^a$ is the exact minimal polynomial of w^ϑ .

This formulation is ideal for Kummer, norm, and ramification investigations because all redundant proper powers have been removed and the algebraic degree is known exactly.

10.4 Exact growth dichotomy

The repository already formalizes the cumulative subquadratic equivalence (14). The exact block formula gives several immediate variants.

Proposition 10.1 (Block-growth reformulations). *Conjecture 3.1 is equivalent to each of the following:*

1. the dyadic block counts $B(T) = \#(\mathcal{C} \cap [2^T, 2^{T+1}))$ are bounded;
2. $B(T) = o(T)$;
3. $\liminf_{T \rightarrow \infty} B(T)/T = 0$;
4. $B(T) = 1$ for every $T \in \mathbb{N}_0$.

Proof. If the conjecture holds, every block contains only 2^T , so all four properties hold. If it fails, (12) gives $B(T) = 1 + \lfloor (T+1)/\beta \rfloor$, so $B(T)/T \rightarrow 1/\beta > 0$ and none of the first three properties holds. The fourth is visibly equivalent to the candidate statement. $\square$

This is useful because a future analytic theorem need not classify candidates individually. Any uniform $o(T)$ bound in the T th dyadic block would prove the conjecture.

10.5 Compact synchronized S -integral orbit

The orbit (6.5) yields another equivalence.

Proposition 10.2 (Compact-orbit reformulation). *The conjecture fails if and only if there exist integers $M, A > 1$ and an irrational $\beta > 0$ such that, for every $k \geq 1$, the point*

$$\left(\frac{M^k}{2^{\lfloor k\beta \rfloor}}, \frac{A^k}{3^{\lfloor k\beta \rfloor}} \right) \quad (45)$$

lies on the compact arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}.$$

Under these conditions the orbit is dense in $\mathcal{A}$.

Proof. Failure supplies the least generator β , with $M = 2^\beta$ and $A = 3^\beta$; then (45) is exactly $(2^{\{k\beta\}}, 3^{\{k\beta\}})$, and density follows from irrational rotation.

Conversely, arc membership gives $t_k \in [0, 1)$ such that

$$k \log_2 M - \lfloor k\beta \rfloor = t_k = k \log_3 A - \lfloor k\beta \rfloor.$$

Thus $\log_2 M = \log_3 A =: \gamma$, and $0 \leq k\gamma - \lfloor k\beta \rfloor < 1$ for every k . Hence $\lfloor k\gamma \rfloor = \lfloor k\beta \rfloor$, so $|k(\gamma - \beta)| < 1$ for every k ; therefore $\gamma = \beta$. Consequently $M = 2^\beta$ and $A = 3^\beta$ are integers while β is nonintegral. The same identity now shows that the orbit equals the irrational-rotation orbit and is dense. $\square$

This formulation isolates the synchronized-denominator feature that a specialized S -integral counting theorem would need to exploit.

10.6 Two-dimensional exponential grid

Under failure, every solution coordinate (n, k) gives

$$m_{n,k} = 2^n M^k, \quad m_{n,k}^\vartheta = 3^n A^k.$$

For every mixed exponent $a + b\vartheta$,

$$m_{n,k}^{a+b\vartheta} = (2^a 3^b)^n (M^a A^b)^k. \quad (46)$$

Thus the candidate evaluation matrix factors through a two-dimensional row lattice and a two-dimensional column lattice. The conjecture can be viewed as the assertion that this integer exponential grid cannot exist unless (M, A) is a common integral power of $(2, 3)$. This is the most natural starting point for a bespoke 2×2 interpolation determinant that uses more than arbitrary-node geometry.

11 Promising future research directions

The priorities below are ordered by how directly they address the quantitative deficit in [section 9](#).

11.1 Priority 1: p -adic amplification of semigroup determinants

The determinant

$$D = \det(m_i^{a_j} (m_i^\vartheta)^{b_j})$$

is a positive integer. Under the exact monoid,

$$m_i = 2^{n_i} M^{k_i}, \quad m_i^\vartheta = 3^{n_i} A^{k_i}.$$

Every entry therefore has explicit valuations

$$\begin{aligned} v_2(\text{entry}_{ij}) &= a_j(n_i + v_2(M) k_i) + b_j v_2(A) k_i, \\ v_3(\text{entry}_{ij}) &= a_j v_3(M) k_i + b_j(n_i + v_3(A) k_i). \end{aligned}$$

A naive row or column factor is lost because the exponent set contains $(0, 0)$. More subtle min-plus determinant estimates may nevertheless force divisibility after suitable row/column finite-difference operations. The exact affine primitivity and the condition $\gcd(d, e, |b-a|) = 1$ should be incorporated rather than discarded.

A concrete target is:

Question 11.1. *Can one choose $O(\log X)$ candidate rows and mixed-monomial columns so that the resulting nonzero determinant has*

$$\log |D|_p^{-1} \gg (\log X)^2$$

for $p = 2$ or 3 , while the archimedean estimate is $\log |D|_\infty = o((\log X)^2)$?

A positive answer would activate the product formula or simply the divisibility lower bound $|D| \geq p^L$ and could erase the missing logarithm.

11.2 Priority 2: exploit the full (n, k) grid, not sorted magnitudes

The arbitrary-node argument forgets the canonical coordinates and retains only the sorted integers m_i . Equation (46) shows that rectangular subsets of rows and columns produce structured bivariate exponential matrices, often Kronecker-like. This is exactly the balanced 2×2 setting where ordinary six-exponentials estimates stop. Integrality, positivity, and the one-depth-zero normalization may create a margin absent from the general four-exponentials problem.

The right experiment is not another large generic determinant. It is an optimized determinant or minor selected from triangular coordinate regions

$$n + k\beta \leq T, \quad a + b\vartheta \leq L,$$

with exact accounting of archimedean size, zero estimates, and 2-/3-adic valuations. Computer algebra can search small shapes for unexpectedly large systematic factors before a general theorem is attempted.

11.3 Priority 3: denominator-sensitive counting on the compact arc

The compact orbit formulation asks for infinitely many points on $y = x^\vartheta$ in $[1, 2) \times [1, 3)$ whose first denominator is a power of 2, whose second denominator is a power of 3, and whose denominator exponents are synchronized by the same integer $\lfloor k\beta \rfloor$.

Standard rational-point counting treats denominator size but usually not this support and synchronization. A theorem of the following flavor would suffice:

On a compact transcendental arc with nonzero curvature, the number of points of logarithmic height at most H whose coordinate denominators are supported on two prescribed disjoint prime sets and have a common exponent parameter is $o(H)$.

The failed-conjecture orbit has $\asymp H$ such points. Butler’s connection between Wilkie-type statements and real exponential conjectures suggests that this direction is genuinely close to the transcendence frontier rather than a routine application of Pila–Wilkie.

11.4 Priority 4: mixed archimedean–radical invariants

The normalized radical $E = w^\vartheta$ has exact degree d and binomial minimal polynomial. Promising questions include:

- Can the logarithmic relation $\log E \log 2 - \log w \log 3 = 0$ force an impossible regulator or height relation in the Kummer field $\mathbb{Q}(E, \zeta_d)$?
- Does simultaneous normalization at both outputs constrain d and e through field intersections or discriminants beyond the known gcd condition?
- Can one use all complex branches $\log(\zeta_d^j E) = \log E + 2\pi i j/d$ in a multi-logarithm determinant whose algebraic entries have controlled height?
- Is there a useful p -adic logarithm branch compatible with the real identity after raising to the normalized degree?

A warning is essential: merely taking norms or conjugates of $X^d - c/3^a$ reproduces known relations. The new invariant must mix conjugation with the special logarithmic direction.

11.5 Priority 5: force a third algebraic exponential

The denominator-controlled theorem in `RationalThirdBase.lean` gives a precise gateway. For an auxiliary integer q containing a prime outside $\{2, 3\}$, it is enough in several forms to prove that q^x is rational with denominator supported by the known outputs. Then monomial injectivity and the three-base determinant force $x \in \mathbb{Q}$, hence $x \in \mathbb{Z}$.

Research should therefore search for natural expressions in M, A whose logarithms introduce a new prime direction but whose x th powers simplify. Straight monomials cannot work; possible candidates would involve norms, resultants, or algebraic combinations arising from the primitive radical field.

11.6 Priority 6: computational reconnaissance with exact certificates

The finite scan through 2^{20} is evidence, not a path to infinity. Computation can still be valuable in three focused roles:

1. search candidate determinant shapes for large systematic 2- or 3-adic valuations;
2. enumerate small primitive radical patterns (w, d, a, c) and record local obstructions, with exact rational inequalities rather than floating-point rejection;
3. compress finite-check certificates through continued-fraction tiers or reusable interval lemmas so that larger ranges can be replayed by the Lean kernel.

The goal should be pattern discovery and theorem generation, not merely extending a decimal barrier.

12 Lean formalization blueprint for the new results

The new arguments are deliberately organized into reusable lemmas. A possible module layout under `ExponentialIdentities/TwoBaseIntegerExponent/` is as follows.

12.1 `ArbitraryNodeDeterminant.lean`

This module should formalize the ordinary-power determinant and Theorem 7.1. The main dependencies are:

1. a determinant/divided-difference identity for the matrix with columns $1, x, \dots, x^{r-1}, f(x)$;
2. a generalized mean-value theorem for divided differences at distinct ordered nodes;
3. the existing derivative tower `rpowDerivativeFamily` and `fallingRpowCoeff` from `HigherDifference.lean`;
4. the integer-entry determinant lower bound already used elsewhere in the project.

A target statement, schematically, is:

```
theorem vandermonde_lower_bound_of_twoBaseNaturalCandidates
  {r : ℕ} (hr : 2 ≤ r)
  (m : Fin (r + 1) → ℕ)
  (hstrict : StrictMono m)
  (hcand : forall i, TwoBaseNaturalCandidate (m i)) :
  (r.factorial : ℝ) / |fallingRpowCoeff logThreeDivLogTwo r| *
    (m 0 : ℝ) ^ ((r : ℝ) - logThreeDivLogTwo)
    ≤ productOverPairs fun i j => ((m j : ℝ) - m i)
```

Listing 1: Proposed arbitrary-node theorem signature.

It may be easier first to state the theorem for a strictly increasing `Finset` or a `Fin`-indexed sequence with an explicit pair product.

12.2 `GeneralizedVandermonde.lean`

The positivity theorem is useful beyond this conjecture. Two formalization routes are available.

Wronskian route. Prove the determinant formula (24); invoke or develop the criterion that positive initial Wronskians imply an extended complete Chebyshev system and positive ordered evaluation determinants.

Zero-count route. After the substitution $x = e^t$, prove by induction that a nonzero exponential polynomial with $r + 1$ distinct real exponents has at most r zeros counting multiplicity. Deduce nonvanishing of the evaluation determinant, then determine its sign by continuity and a coalescing-node limit.

The zero-count route may require more real-analysis infrastructure, while the Wronskian route may require adding a small Chebyshev-system API. For an initial project-specific proof, one can prove only nonvanishing and then compare signs through total positivity of the exponential kernel e^{st} .

12.3 MixedExponentSemigroup.lean

Define a pair-indexed exponent

```
def mixedExponent (ab : N x N) : R :=
  ab.1 + ab.2 * logThreeDivLogTwo
```

and prove:

- injectivity from irrationality of ϑ ;
- integrality of $m^{a+b\vartheta} = m^a(m^\vartheta)^b$ for candidates;
- positivity and integer-valuedness of the corresponding generalized determinant;
- finite ordering of any chosen list of pairs by their real exponents.

For the quantitative theorem, it is unnecessary initially to define the globally sorted semigroup. One may accept an arbitrary strictly increasing finite exponent list and a proof that every exponent has a pair representation.

12.4 SemigroupDividedDifferenceBound.lean

Formalize Lemma 8.3 and the strip bound (30). The main analytic statement should be a bound on every Newton row:

```
abs (dividedDiff (fun x : R => x ^ lambda) nodesPrefix)
  <= 2 ^ (2 * lambda + i) * N ^ (lambda - i)
```

under $N \leq x \leq 2N$ and $\lambda \geq 0$.

The determinant estimate then follows from `Matrix.det_apply` or a suitable Leibniz-bound lemma. It would be worth adding a generic API:

if every entry in row i , column j is bounded by a separable quantity $A_i B_j$, then the determinant is at most $(r+1)! \prod_i A_i \prod_j B_j$.

12.5 SemigroupCounting.lean

For the explicit logarithmic-square corollary, the full asymptotic (33) is optional. Formalize the square-box construction from Lemma 8.5:

- the $(q+1)^2$ pairs in $[0, q]^2$ give distinct mixed exponents;
- every such exponent is below $3q$ using $1 < \vartheta < 2$;
- hence the r th order statistic is at most $6\sqrt{r}$;

- sum and divide to obtain $\Sigma_r/R_r \leq 12/\sqrt{r}$.

Avoiding real-valued order statistics may be easier if the theorem chooses any $r + 1$ pairs from a square box and bounds their total weight directly. For example, take the first $r + 1$ pairs from a fixed enumeration of $[0, q]^2$; after sorting by exponent, their sum is at most $3q(r + 1)$. This proves the same coarse determinant exponent without constructing the canonical ordered semigroup.

12.6 SemigroupDyadicBound.lean

Combine the span theorem with the consecutive-window summation argument. The final theorem could state:

```
theorem card_twoBaseNaturalCandidates_Icc_le_log_sq (X : N) (hX : 1 <= X) :
  ((Finset.Icc X (2 * X)).filter TwoBaseNaturalCandidate).card
  <= Nat.ceil (10000 * max 1 (Real.log X) ^ 2)
```

Listing 2: Proposed explicit dyadic bound.

A cleaner all-natural statement may replace the real logarithm by the binary ceiling logarithm, obtaining a larger but easier constant times $\text{Nat.clog } 2 \ X \ ^2$. Since the repository already has logarithmic-square counting infrastructure in `MultiplicativeRank.lean`, its finite-set definitions and cardinality lemmas can be reused while keeping the proof mechanism independent.

12.7 Recommended proof-audit discipline

Each new headline theorem should be included in the aggregate import and checked with `#print axioms`. The report status should remain **paper proof in this report** until the source is committed and a full build succeeds. Numerical constants in tables are explanatory only; the formal theorem should use exact expressions or safe rational constants such as $1/24$ and 10000 .

13 A more detailed comparison with existing ProveIt methods

13.1 Prime-valuation rank versus interpolation geometry

The module `MultiplicativeRank.lean` proves that any three candidates have linearly dependent prime-exponent vectors of rank at most two. It does so by a three-base/six-exponentials determinant: if three candidate bases were multiplicatively independent, the integrality of all three ϑ th powers would force ϑ rational, contradicting its irrationality. A finite candidate set therefore lives in a rational plane in the free vector space on primes. Choosing two prime coordinates injectively encodes the set and yields an explicit logarithmic-square count.

The semigroup determinant proves a logarithmic-square dyadic bound by a different mechanism:

Prime-valuation method	Semigroup determinant method
Uses prime factorization of candidate bases	Uses only integer values on $y = x^\vartheta$
Global multiplicative rank at most two	Local total positivity and interpolation
Invokes a six-exponentials consequence	Uses elementary determinant integrality once ϑ is fixed
Produces coordinates in two prime directions	Produces repulsion for arbitrary ordered nodes
Naturally global in magnitude	Naturally local in additive intervals
Already kernel-verified	Paper proof proposed for formalization

Their agreement at logarithmic-square scale is suggestive. Both see a two-dimensional structure: prime-exponent rank two on one side, mixed-exponent lattice dimension two on the other. A future proof may need to combine these dual lattices rather than refine either in isolation.

13.2 Exact conditional rank one over a two-generator monoid

Conditional on failure, the candidate monoid is not merely rank at most two; it is exactly free on 2 and M . In prime-valuation space its points form the nonnegative lattice generated by the vectors of 2 and M . In mixed-exponent evaluation space, the values factor as in (46). This exact grid is much stronger than the unconditional rank theorem, but all presently known consequences remain compatible with it.

The strongest likely synthesis is a determinant whose rows use primitive-generator coordinates and whose columns use mixed exponents. Such a matrix is simultaneously:

- integral;
- totally positive after ordering by magnitude;
- explicitly factored into powers of 2, 3, M , A ;
- constrained by affine primitivity and simultaneous output normalization.

No current module appears to combine all four properties.

13.3 Finite differences as confluent determinants

Forward differences correspond to the equally spaced specialization of interpolation, while Wronskians correspond to confluent nodes. The new determinant hierarchy therefore provides a single conceptual envelope for:

- second-, third-, and arbitrary-order finite differences;
- arbitrary-node ordinary-power determinants;
- mixed-monomial generalized Vandermonde determinants;
- possible confluent determinants using derivatives or repeated coordinate directions.

A useful formalization strategy is to build a general interpolation-determinant layer, then recover `HigherDifference.lean` as a specialization. This would reduce duplicated analysis and make experimental determinant choices easier to certify.

14 Conclusions

The Alaoglu–Erdős conjecture remains unresolved here. That is not for lack of structural information. Conditional on one counterexample, the pinned ProveIt corpus determines the entire solution set, candidate monoid, exact block counts, asymptotic population, rotation statistics, primitive odd core, rational-power index, radical degree, and simultaneous output normalization. The surviving obstruction is exceptionally concentrated:

$$\text{exclude } w^{\log_2 3} \in \mathbb{Z}[1/3] \text{ for every primitive odd } w > 1.$$

The new arbitrary-node determinant proves that candidate lattice points cannot cluster freely; all Vandermonde products obey a power-law lower bound. The mixed-exponent semigroup then gives arbitrarily high-order positive integer determinants and a repulsion exponent tending to one. It yields a second, conceptually independent logarithmic-square counting theorem.

The most valuable outcome is the quantified failure point. Pure archimedean interpolation requires about $(\log X)^2$ nodes to operate at scale X , whereas the exact conditional monoid provides only $\log X$. This makes future work more focused. Improving constants or extending finite searches will not close the gap. The plausible routes are those capable of supplying one additional logarithm of determinant strength: 2- or 3-adic divisibility, a synchronized-denominator counting theorem on the compact exponential arc, or a mechanism that forces a third algebraically independent exponential direction.

The new results are sufficiently elementary and modular to formalize now. Even if the full conjecture ultimately requires a major transcendence theorem, adding arbitrary-node and semigroup-determinant APIs to ProveIt would sharpen the machine-checked frontier and provide the right experimental framework for the next attack.

A Detailed proof of the generalized Vandermonde sign

This appendix expands Lemma 8.1 to make its analytic input fully transparent.

Let $0 \leq \lambda_0 < \dots < \lambda_r$ and define $f_j(x) = x^{\lambda_j}$ on $(0, \infty)$. For $k \leq r$,

$$f_j^{(k)}(x) = (\lambda_j)_{\underline{k}} x^{\lambda_j - k}.$$

The k th initial Wronskian is

$$\begin{aligned} W_k(x) &= \det(f_j^{(i)}(x))_{0 \leq i, j \leq k} \\ &= x^{\sum_{j=0}^k \lambda_j - k(k+1)/2} \det((\lambda_j)_{\underline{i}})_{0 \leq i, j \leq k}. \end{aligned}$$

The polynomial $(z)_{\underline{i}}$ is monic of degree i . Replacing the monomial basis $1, z, \dots, z^k$ by the falling-factorial basis is unit lower-triangular. Therefore

$$\det((\lambda_j)_{\underline{i}})_{i, j=0}^k = \prod_{0 \leq i < j \leq k} (\lambda_j - \lambda_i) > 0.$$

Hence every initial Wronskian is positive.

One standard characterization of extended complete Chebyshev systems now implies that for $x_0 < \cdots < x_r$,

$$\det(f_j(x_i))_{i,j=0}^r > 0.$$

To avoid using that characterization as a black box, proceed inductively. Suppose a nontrivial combination

$$F(x) = \sum_{j=0}^r c_j x^{\lambda_j}$$

has $r + 1$ distinct positive zeros. Put $x = e^t$ and factor the lowest exponential:

$$e^{-\lambda_0 t} F(e^t) = c_0 + \sum_{j=1}^r c_j e^{(\lambda_j - \lambda_0)t}.$$

Rolle's theorem gives at least r zeros of its derivative, an exponential polynomial with at most r terms. Induction gives a contradiction unless all coefficients vanish. Thus the evaluation determinant is nonzero. Its sign cannot change on the connected chamber $x_0 < \cdots < x_r$. Let the nodes coalesce as $x_i = x + i\varepsilon$ and divide by the positive ordinary Vandermonde. As $\varepsilon \downarrow 0$, the quotient tends to $W_r(x) / \prod_{k=0}^r k! > 0$. Hence the original determinant is positive everywhere in the ordered chamber.

B Numerical constants for arbitrary-node repulsion

For reference,

$$\vartheta = 1.58496250072115618145\dots, \quad \vartheta(\vartheta - 1) = 0.92714362797110482756\dots$$

The three-point span constant is

$$\left(\frac{8}{\vartheta(\vartheta - 1)} \right)^{1/3} = 2.051072395661874\dots$$

For the ordinary-power order r , the raw span constant and exponent are

$$C_r = \left(\frac{r!}{|(\vartheta)_{\underline{r}}|} \right)^{2/(r(r+1))}, \quad \alpha_r = \frac{2(r - \vartheta)}{r(r + 1)}.$$

The continuous critical point for α_r solves

$$-r^2 + 2\vartheta r + \vartheta = 0,$$

so

$$r_* = \vartheta + \sqrt{\vartheta^2 + \vartheta} = 3.6090841940\dots,$$

confirming that $r = 4$ is the optimal integer order.

For the semigroup hierarchy, the first exponents and their pair labels are:

Index	Pair (a, b)	Exponent $a + b\vartheta$
0	$(0, 0)$	0
1	$(1, 0)$	1
2	$(0, 1)$	ϑ
3	$(2, 0)$	2
4	$(1, 1)$	$1 + \vartheta$
5	$(3, 0)$	3
6	$(0, 2)$	2ϑ
7	$(2, 1)$	$2 + \vartheta$
8	$(4, 0)$	4
9	$(1, 2)$	$1 + 2\vartheta$
10	$(3, 1)$	$3 + \vartheta$
11	$(0, 3)$	3ϑ
12	$(5, 0)$	5

C Repository theorem inventory used in this report

The following modules on the aggregate import surface were especially relevant.

Module	Role
<code>TwoBaseIntegerExponent.lean</code>	Basic definitions, fractional-part gaps, candidate exponent, and low-order curvature.
<code>FiniteCheck4096.lean</code>	Exact natural-arithmetic certificates excluding nonintegral candidates below 4096.
<code>Transcendence.lean</code>	Gelfond–Schneider dichotomy; nonintegral solutions and ϑ are transcendental.
<code>FourExponentialsReduction.lean</code>	Rational linear independence of $(1, x)$ and $(\log 2, \log 3)$; four exponentials implies the conjecture.
<code>Localization.lean</code>	Equivalence with $u^\vartheta \notin \mathbb{Z}[1/3]$ for odd $u > 1$.
<code>PrimitiveRadical.lean</code>	Reduction to power-primitive odd bases.
<code>RadicalDegree.lean</code>	Exact irreducible binomial and algebraic degree for the least rational power.
<code>PrimitiveOddCore.lean</code>	Canonical common odd core of all hypothetical candidates.
<code>RationalPowerIndex.lean</code>	Cyclic subgroup of rational powers and least index.
<code>PrimitiveGenerator.lean</code>	Canonical least generator and exact free solution/candidate monoids.
<code>ExactCounting.lean</code>	Exact unit-block and dyadic-block parametrization and cardinalities.
<code>QuadraticAsymptotic.lean</code>	Sharp cumulative floor-sum bounds, quadratic limit, and growth equivalence.
<code>BlockWeyl.lean</code>	Vanishing Fourier modes for exact block orbit segments.
<code>OutputNormalization.lean</code>	Simultaneous primitive output decompositions and gcd constraints.
<code>RationalThirdBase.lean</code>	Third-base rationality with controlled denominator support.
<code>ThreeSmooth.lean</code>	Classification of auxiliary bases supported on 2 and 3.

Module	Role
<code>HigherDifference.lean</code>	Arbitrary-order forward-difference mean value and progression obstruction.
<code>MultiplicativeRank.lean</code>	Rank at most two for candidate prime-exponent vectors and logarithmic-square counting.

D Suggested experimental determinant families

Before developing a full p -adic theory, several finite experiments can test whether the required amplification exists.

D.1 Rectangular row and column boxes

Choose rows (n, k) with $0 \leq n < N_0$, $0 \leq k < K_0$ and columns (a, b) with $0 \leq a < A_0$, $0 \leq b < B_0$. The entry is

$$(2^a 3^b)^n (M^a A^b)^k.$$

For complete rectangles, the matrix is a rearranged Kronecker product only when the row and column cardinalities match in a compatible way. Its determinant or maximal minors can be factored symbolically for small boxes. Record valuations at 2 and 3 as polynomials in $a_0 = v_2(M)$ and $b_0 = v_3(A)$.

D.2 Triangular boxes adapted to height

Use

$$n + k\beta \leq T, \quad a + b\vartheta \leq L.$$

These are the natural row and column shapes from the exact counting theorem and mixed-exponent semigroup. Search for square minors with minimal archimedean weight and maximal forced valuation. The linear boundary slopes are irrational, so floor-sum fluctuations may prevent cancellation patterns present in rectangles.

D.3 Finite-difference row operations

Order rows by n for fixed k , or by k for fixed n , and apply repeated difference operators. Differences introduce factors such as

$$2^a 3^b - 1, \quad M^a A^b - 1.$$

These may have systematic 2- or 3-adic valuations for carefully selected column pairs. Lifting-the-exponent lemmas can then provide exact lower bounds. The risk is that the archimedean size of the same factors grows too quickly; experiments should track both sides.

D.4 Confluent columns

Instead of only distinct mixed monomials, consider derivatives with respect to an auxiliary continuous exponent parameter before specializing. Confluent generalized Vandermonde matrices introduce logarithmic factors, which are not integral, but common-denominator or linear-form constructions may restore arithmetic control. This is closer to Laurent’s interpolation determinant method and may reduce total exponent weight.

E Bibliographic and repository references

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